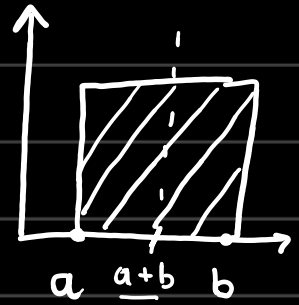


For a uniform distribution:

$$p(x) = \begin{cases} 1/(b-a) & ; a \leq x \leq b \\ 0 & ; \text{otherwise} \end{cases}$$

$$E[x] = \int_a^b x \cdot \frac{1}{b-a} dx = \frac{a+b}{2}$$



$$E[x^2] = \frac{b^3 - a^3}{3(b-a)} \quad \left| \quad \text{var}(x) = E[(x - \mu)^2] \right.$$

(second moment)

$$\begin{aligned} \text{var}(x) &= E[(x - \mu)^2] \\ &= \int_a^b \left(x - \left(\frac{a+b}{2}\right)\right)^2 \cdot \frac{1}{b-a} dx \end{aligned}$$

$$= \frac{1}{b-a} \left[\int_a^b \left(x^2 - (a+b)x + \frac{(a+b)^2}{4}\right) dx \right]$$

$$1) \int_a^b x^2 \cdot dx = \frac{b^3 - a^3}{3}$$

$$2) \int_a^b x \cdot dx = \frac{b^2 - a^2}{2}$$

$$3) \int_a^b 1 \cdot dx = b - a$$

$$\Rightarrow = \frac{1}{b-a} \left[\frac{b^3 - a^3}{3} - (a+b) \left(\frac{b^2 - a^2}{2} \right) + \frac{(a+b)^2}{4} (b-a) \right]$$

$$= \frac{1}{b-a} \left[\frac{(b-a)(b^2 + ab + a^2)}{3} - \frac{(a+b)(b+a)(b-a)}{2} + \frac{(a+b)^2(b-a)}{4} \right]$$

$$= \frac{b^2 + ab + a^2}{3} - \frac{(a+b)^2}{2} + \frac{(a+b)^2}{4}$$



$$\begin{aligned}
&= \frac{b^4 + ab + a^4}{3} - \frac{(a+b)^2}{4} \\
&= \frac{4b^4 + 4ab + 4a^2 - 3a^2 - 6ab - 3b^2}{12} \\
&= \frac{b^2 - 2ab + a^2}{12} = \frac{(b-a)^2}{12}
\end{aligned}$$

For Gaussian (Normal) Distribution

$$b_x(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$E[x] = \int_{-\infty}^{\infty} x \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

(Expected value)

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} x \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

replace $\frac{x-\mu}{\sqrt{2}\sigma} = y \Rightarrow x-\mu = y \cdot \sqrt{2}\sigma$
 $dx = \sqrt{2}\sigma dy$

$$= \frac{1}{\cancel{\sigma\sqrt{2\pi}}} \int_{-\infty}^{\infty} (\sqrt{2}\sigma y + \mu) \cdot e^{-y^2} \cdot dy \cdot \cancel{\sqrt{2}\sigma}$$

$$= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} \sqrt{2}\sigma y \cdot e^{-y^2} \cdot dy + \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} \mu \cdot e^{-y^2} dy$$

$$= \frac{\sqrt{2}\sigma}{\sqrt{\pi}} \int_{-\infty}^{\infty} y \cdot e^{-y^2} \cdot dy + \frac{\mu}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-y^2} dy$$

odd

$$= \frac{2\mu}{\sqrt{\pi}} \int_0^{\infty} e^{-y^2} dy = \frac{2\mu}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = \mu$$



$$\begin{aligned}
 \text{var}(x) &= \int_{-\infty}^{\infty} (x-\mu)^2 \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \\
 &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} 2\sigma^2 y^2 \cdot e^{-y^2} \sqrt{2\pi} dy \\
 &= \frac{2\sigma^2}{\sqrt{\pi}} \int_{-\infty}^{\infty} y^2 \cdot e^{-y^2} dy
 \end{aligned}$$

$$\text{let } y^2 = t \Rightarrow 2y dy = dt$$

$$\begin{aligned}
 &= \frac{2\sigma^2}{\sqrt{\pi}} \int_{-\infty}^{\infty} t \cdot e^{-t} \frac{dt}{2\sqrt{t}} = \frac{\sigma^2}{\sqrt{\pi}} \int_{-\infty}^{\infty} t \cdot \frac{e^{-t}}{t^{1/2}} dt \\
 &= \frac{\sigma^2}{\sqrt{\pi}} \int_{-\infty}^{\infty} t^{1/2} e^{-t} dt \\
 &= \frac{\sigma^2}{\sqrt{\pi}} \cdot \sqrt{\pi} = \sigma^2
 \end{aligned}$$

Derivation

$$\text{To prove: } \int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

① Square the integral

$$I^2 = \int_{-\infty}^{\infty} e^{-x^2} dx \cdot \int_{-\infty}^{\infty} e^{-y^2} dy$$

$$\text{use } \left(\int f(x) dx \right) \left(\int g(y) dy \right) = \iint f(x) g(y) dx dy$$

$$I^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-x^2} \cdot e^{-y^2} dx \cdot dy = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy$$

Polar Co-ordinates

- $x = r \cos \theta$
- $y = r \sin \theta$
- $x^2 + y^2 = r^2$
- Area element: $dx dy = r dr d\theta$
- $(x, y) \in \mathbb{R}^2$
- $r \in [0, \infty)$
- $\theta \in [0, 2\pi)$



$$\text{So, } I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r \, dr \, d\theta$$

$$\cdot \int_0^{\infty} e^{-r^2} r \, dr$$

$$\text{let } u = r^2 \Rightarrow du = 2r \, dr$$

$$\Rightarrow r \, dr = du/2$$

$$\text{when } r = 0, u = 0$$

$$\text{when } r \rightarrow \infty, u \rightarrow \infty$$

So :

$$\int_0^{\infty} e^{-r^2} r \, dr = \int_0^{\infty} e^{-u} \cdot \frac{1}{2} du = \frac{1}{2} \int_0^{\infty} e^{-u} du$$

$$= \frac{1}{2} [e^{-u}]_0^{\infty} = \frac{1}{2} (0 - (-1)) = \frac{1}{2} \times 1 = \frac{1}{2}$$

$$\Rightarrow I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r \, dr \, d\theta = \int_0^{2\pi} \frac{1}{2} d\theta$$

$$= \frac{1}{2} \cdot (2\pi) = \pi$$

$$\text{So, } I^2 = \pi \Rightarrow \boxed{I = \sqrt{\pi}}$$

Hello

